

H2A.Z deposition at meiotic prophase I underlies homologous recombination and pachytene genome activation during male meiosis

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Abstract

Accurate meiotic progression is important for gamete formation and the generation of genetic diversity. However, little is known about the identity of chromatin regulators that underlie mammalian meiosis *in vivo*. Here, we identify the multifaceted functions of the chromatin remodeler Znhit1 in governing meiosis. We observe a gradual increase in *Znhit1* expression during the meiotic prophase. Znhit1 deficiency in spermatocytes results in arrested pachytene development, impaired DNA double-strand break repair, and defective homologous recombination. Single-cell RNA sequencing and transcriptome analysis reveal that Znhit1 loss downregulates the transcription of pachytene genome activation (PGA) genes globally. Chromatin immunoprecipitation data show that Znhit1 is needed for the incorporation of the histone variant H2A.Z into pachytene chromatin. Moreover, we find that H2A.Z cooperates with the transcription factor A-MYB to co-bind DNA elements and control enhancer activity. Our findings provide insights into the regulatory mechanisms governing meiotic progression and highlight Znhit1 as a critical regulator of meiotic recombination and PGA.

eLife assessment

This study shows that *Znhit1*, a regulator of chromatin and of the histone variant H2A.Z, is required for progression through meiotic prophase. It is an **important** observation that describes the role of epigenetics and gene expression during meiosis. The analysis is based on complementary approaches at the cytological, single-cell, and genomic levels that provide **solid** evidence for the role of *Znhit1* in the control of gene expression and in the loading of H2A.Z in mouse spermatocytes.

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Introduction

Meiosis is a fundamental and conserved process that plays a crucial role in gamete formation and the generation of genetic diversity. By undergoing one round of DNA replication and two rounds of cell division, the meiotic process ensures the production of haploid cells, each with a unique combination of genetic materials (Handel and Schimenti, 2010; Lascarez-Lagunas et al., 2020). Any disruptions to the normal progression of meiosis would have significant consequences, such as aneuploidy, infertility, spontaneous abortion, and congenital diseases (Hassold and Hunt, 2001).

Therefore, understanding the molecular mechanisms underlying meiosis will provide valuable insights for the diagnosis and treatment of reproductive and developmental diseases.

Multiple chromosomal events occur during meiotic prophase I, including homologous chromosome pairing and synapsis, DNA double-strand break (DSB) formation and repair, and homologous recombination (HR) (Baudat et al., 2013; Hunter, 2015; Zickler and Kleckner, 2023). In this process, homolog pairing and synapsis coincide with Spo11- mediated genome-wide formation of DSBs (Bergerat et al., 1997; Keeney et al., 1997).

Following the formation of DSBs, the DNA undergoes resection, leading to the generation of single-stranded DNA overhangs. These overhangs are then coated by RPA, DMC1, and RAD51, facilitating the production of recombination intermediates.

Subsequently, these intermediates are processed and resolved, ultimately forming either meiotic crossovers or non-crossovers (Gray and Cohen, 2016; San Filippo et al., 2008; Symington, 2014). The tightly coordinated timing and spatial arrangement of these meiotic events are of utmost importance for proper germ cell development.

There is transcriptional awakening during male meiotic prophase I, in which the quiescent meiotic genome becomes transcriptionally active during the zygotene-to-pachytene transition (Ernst et al., 2019; Green et al., 2018; Rabbani et al., 2022; Turner, 2015). This process, referred to as pachytene genome activation (PGA), is responsible for the expression of numerous genes and plays an essential role in controlling meiotic and post- meiotic events. Previous studies have emphasized the significance of the transcription factors (TFs) A-MYB, BRDT, and TCFL5 in activating transcription during pachytene (Alexander et al., 2023; Bolcun-Filas et al., 2011; Cecchini et al., 2023; Gaucher et al., 2012; Li et al., 2013; Maezawa et al., 2020; Manterola et al., 2018; Ozata et al., 2020; Yu et al., 2021). Additionally, extensive chromatin remodeling takes place in spermatocytes during meiotic prophase I, involving histone variant exchange, histone modifications, and high-order genome rearrangement (Kota and Feil, 2010; Wang et al.,

2017 [\[1\]](#); Zheng and Xie, 2019 [\[2\]](#)). It has been known that SETDB1-mediated H3K9 trimethylation (H3K9me3) is required for sex chromosome transcription silencing (a process called meiotic sex chromosome inactivation, MSCI) and male meiotic procession (Hirota et al., 2018 [\[3\]](#)).

However, the specific involvement of chromatin regulators in gene activation on autosomes remains poorly understood.

Zinc finger HIT-type containing 1 (Znhit1), an evolutionarily conserved subunit of the SRCAP chromatin remodeling complex, acts as a key regulator for the histone variant H2A.Z deposition to control gene expression and cell fate determination (Cai et al., 2005 [\[4\]](#); Cuadrado et al., 2007 [\[5\]](#); Feng et al., 2022 [\[6\]](#)). Recent studies have shown that Znhit1 is involved in a wide range of developmental processes, including muscle and lens differentiation, heart development, lung branching, and adult tissue stem cell maintenance (Cuadrado et al., 2010 [\[7\]](#); Lu et al., 2022 [\[8\]](#); Sun et al., 2020 [\[9\]](#); Wei et al., 2022 [\[10\]](#); Xu et al., 2021 [\[11\]](#); Zhao et al., 2019 [\[12\]](#)). In our most recent study, we deleted Znhit1 in early male germ cells and found that Znhit1 is required for the mitosis-to-meiosis transition by controlling Meiosin expression (Sun et al., 2022 [\[13\]](#)). However, the in vivo function of Znhit1 in regulating meiotic progression remains unknown.

To delve into the function of Znhit1 in meiosis, we analyzed Znhit1 expression and observed upregulated Znhit1 expression specifically during meiotic prophase. Here, we examine the role of Znhit1 in multiple meiotic events and show that the deletion of Znhit1 in spermatocytes causes impaired HR and defective PGA, thereby resulting in meiotic developmental arrest. Hence, our study highlights the essential role of Znhit1 in ensuring meiotic progression, implicating its potential as a therapeutic target for meiosis-related diseases.

Results

Znhit1 expression is upregulated during meiotic prophase and Znhit1 knockout in spermatocytes disrupts spermatogenesis

To identify potential chromatin regulators in meiosis, we first queried for chromatin factors highly expressed during the zygotene stage based on published transcription data (Fig S1A and Table S1) (Chen et al., 2018 [\[14\]](#)). Quantitative analysis showed that the chromatin remodeler Znhit1 was among the most highly expressed factors (Figs 1A [\[15\]](#), 1B [\[16\]](#), S1B, and S1C). RNA in situ hybridization validated that *Znhit1* expression in male germ cells decreased after meiotic initiation, followed by a gradual increase in spermatocytes from the zygotene stage to the metaphase stage (Fig 1C [\[17\]](#)). These results suggest that Znhit1 is a potential chromatin regulator of meiotic progression.

To investigate whether Znhit1 regulates meiotic progression, we generated spermatocyte-specific *Znhit1* knockout mice (*Znhit1*^{fl/fl}; *Stra8-cre*, referred to as *Znhit1*-sKO). In the first wave of spermatogenesis, Ng3-negative prospermatogonia directly differentiate into A2 spermatogonia (Law et al., 2019 [\[18\]](#); Rabbani et al., 2022 [\[19\]](#); Yoshida et al., 2006 [\[20\]](#); Yoshida et al., 2004 [\[21\]](#)). With *Stra8-cre* inducing recombination in A1 spermatogonia and preleptotene spermatocytes separately (Lin et al., 2017 [\[22\]](#)), it is feasible to generate mice with spermatocyte-specific gene knockout (Fig 1D [\[23\]](#)). RNA in situ experiments on testis sections confirmed *Znhit1* mRNA deletion in spermatocytes (Figs 1E [\[24\]](#) and S1D). Compared with littermate controls, *Znhit1*-sKO male mice had smaller testes (Fig S1E and S1F). PAS- histological and PNA-fluorescence staining illuminated the absence of round spermatids and elongated spermatids in *Znhit1*-sKO testes (Fig 1F [\[25\]](#) and 1G [\[26\]](#)), indicating that *Znhit1* knockout in spermatocytes disrupted spermatogenesis.

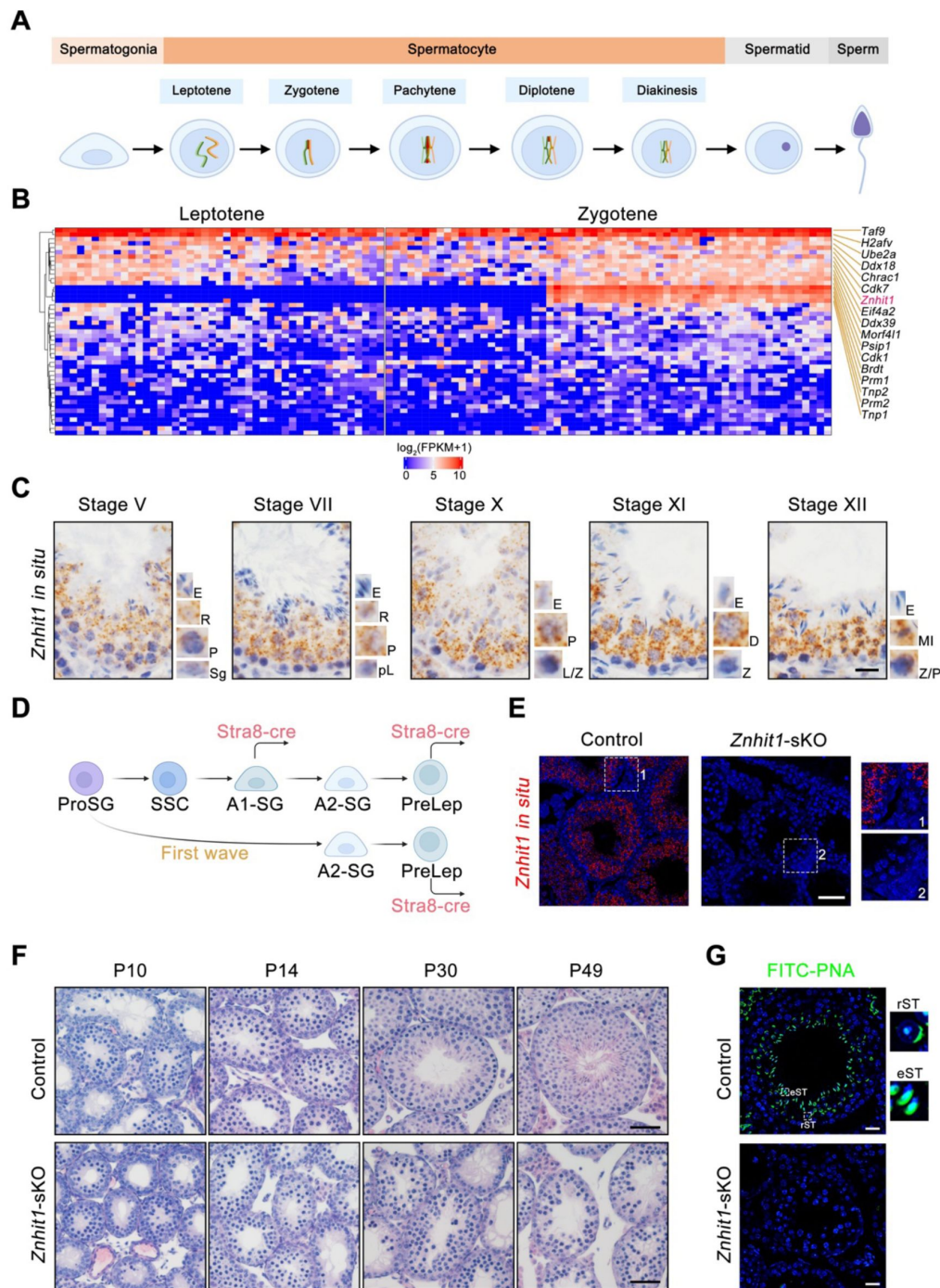


Figure 1

Znhit1 deletion in spermatocytes disrupts spermatogenesis.

(A) Schematic representation of the male germline development in mice. (B) Heatmap showing the clustering of 47 epigenetic factors based on the transcription levels in leptotene and zygotene spermatocytes. (C) *Znhit1* in situ hybridization in postnatal day 60 (P60) testis sections. Scale bar, 20 μm . Sg, spermatogonia; pL, preleptotene; L, leptotene; Z, zygotene; P, pachytene; D, diplotene; MI, metaphase; R, round spermatids; E, elongating spermatids. (D) Schematic diagram showing the onset of *Stra8-cre* expression in the male germline development. (E) *Znhit1* in situ hybridization in P42 testis sections. Scale bar, 50 μm . (F) Histological testicular sections in control or *Znhit1*-sKO testis sections at the indicated times. Scale bar, 50 μm . (G) Immunostaining of PNA in testis sections from control or *Znhit1*-sKO mice at the indicated times. rST, round spermatids; eST, elongating spermatids. Scale bar, 20 μm . All images are representative of $n = 3$ mice per genotype.

Znhit1 deletion leads to meiotic arrest at the pachytene stage

To dissect spermatocyte development in *Znhit1*-sKO testes, we co-stained SYCP3, a marker of primary spermatocytes, and HSPA2, a testis-specific HSP70 family member that accumulates from the pachytene stage onward. As shown in **Figure 2A**, both control and *Znhit1*-sKO testis sections exhibited SYCP3⁺HSPA2⁺ primary spermatocytes.

Phospho-histone H3 (Thr3) (pH3) is a marker of mitotic and meiotic metaphase cells, and *Znhit1* deletion caused the complete absence of pH3⁺ metaphase spermatocytes within the cavity of seminiferous tubules (**Fig 2B**). These results suggest that *Znhit1* deletion causes spermatocyte arrest at meiotic prophase I.

To pinpoint the onset of meiotic failure, we performed immunostaining against SYCP3 in spermatocyte chromosome spreads. The wild-type testes displayed typical spermatocytes at consecutive substages, including leptotene, zygotene, pachytene, diplotene, and diakinesis (**Fig 2C**). However, spermatocytes at the diplotene and diakinesis stages were rarely observed in *Znhit1*-sKO testes (**Fig 2D** and **2E**). We also examined the expression of H1T, a middle pachytene marker, and no difference in H1T expression was observed in *Znhit1*^{-/-} spermatocytes (**Fig S2A**). Moreover, *Znhit1* knockout resulted in increased TUNEL-positive spermatocytes, suggesting that these defective spermatocytes were eliminated by the apoptotic pathway (**Fig 2F** and **2G**). These results indicate that *Znhit1* deletion leads to meiotic arrest at the pachytene stage.

Znhit1 is required for DSB repair and meiotic recombination

To investigate the impact of *Znhit1* on meiotic events, we first performed immunostaining of SYCP1 and SYCP3, the essential components of the synaptonemal complex, to examine whether *Znhit1* regulated homologous synapses. As shown in **Figure 3A**, pachytene nuclei with fully synapsed autosomes were observed in spermatocytes from control and *Znhit1*-sKO testes. During the pachytene stage, the X and Y chromosomes undergo pairing and synapses between the short pseudoautosomal regions (PARs). We found an increase in the percentage of unsynapsed X-Y chromosomes in *Znhit1*-sKO spermatocytes (**Fig 3B**). Thus, deletion of *Znhit1* results in impaired sex chromosomal synapsis.

Next, we examined the programmed formation and repair of DSBs by co-staining SYCP3 and γH2AX, a DSB marker, on spermatocyte chromosome spreads. In both control and *Znhit1*-sKO spermatocytes, γH2AX signals were evident in the leptotene and zygotene stages (**Fig 3C**). However, in *Znhit1*-deficient pachynema, we still observed diffuse γH2AX signals on the autosomes, while in control pachynema, γH2AX signals only accumulated on the X-Y chromosomes. Moreover, we detected the presence of another DSB marker, phospho-ATM (pATM), on the autosomes in *Znhit1*-sKO pachytene spermatocytes (**Fig 3D**). Together, these results demonstrate that *Znhit1* is required for completing DSB repair.

To understand how *Znhit1* regulates DSB repair, we performed an immunostaining analysis using the markers RPA2 and DMC1, the essential factors involved in DSB repair by coating resected DSBs. We found an increase in RPA2 and DMC1 counts in both zygonema and pachynema of *Znhit1*-deficient testes (**Fig 3E**, **3F**, and **3A**), indicating that *Znhit1* deletion delayed recombinational repair, resulting in defective resolution of autosomal DSBs in *Znhit1*^{-/-} pachynema. The formation and repair of DSBs play crucial roles in initiating and facilitating meiotic recombination, respectively. To examine the consequences of unrepaired DSBs on meiotic recombination in *Znhit1* mutants, we performed an immunostaining analysis against MLH1, a marker of meiotic crossover formation. Indeed, we found an almost absence of MLH1 foci in *Znhit1*^{-/-} pachytene spermatocytes compared with controls (**Fig 3G** and **3H**), indicating that the loss of *Znhit1* severely disrupted meiotic crossover formation.

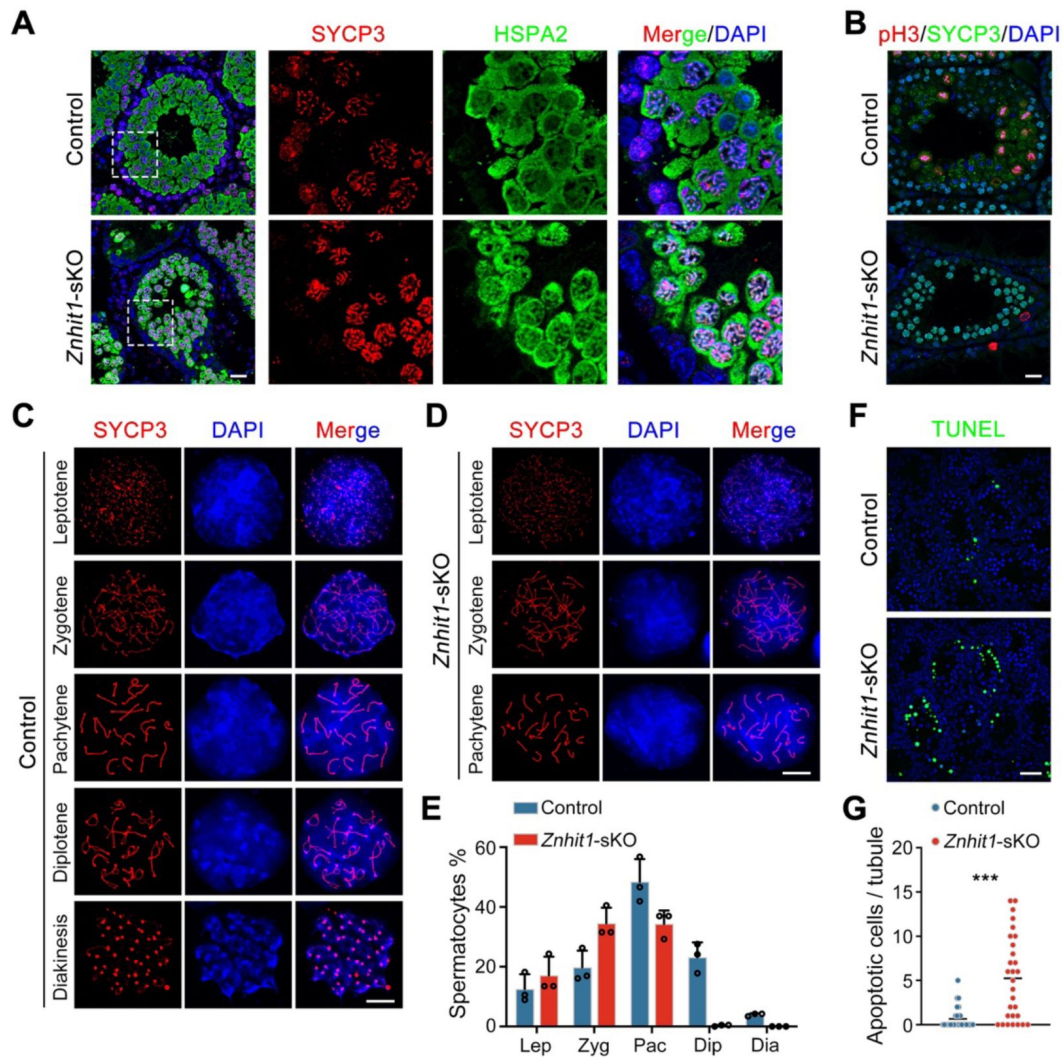


Figure 2

***Znhit1* deletion leads to meiotic pachytene arrest.**

(A) Immunostaining of SYCP3 and HSPA2 in testis sections of P35 control or *Znhit1*-sKO mice. Scale bar, 20 μ m. (B) Immunostaining of SYCP3 and pH3 in testis sections of P35 control or *Znhit1*-sKO mice. Scale bar, 20 μ m. (C-E) Immunostaining of SYCP3 in spermatocyte chromosome spreads of control (C) or *Znhit1*-sKO mice (D). Quantitative data are shown in (E). Scale bar, 10 μ m. (F and G) TUNEL staining in testis sections from P14 control or *Znhit1*-sKO mice. Quantitative data are shown in (G). Scale bar, 40 μ m. All images are representative of $n = 3$ mice per genotype. Data are presented as the mean $\pm$ s.d. *** $p < 0.001$.

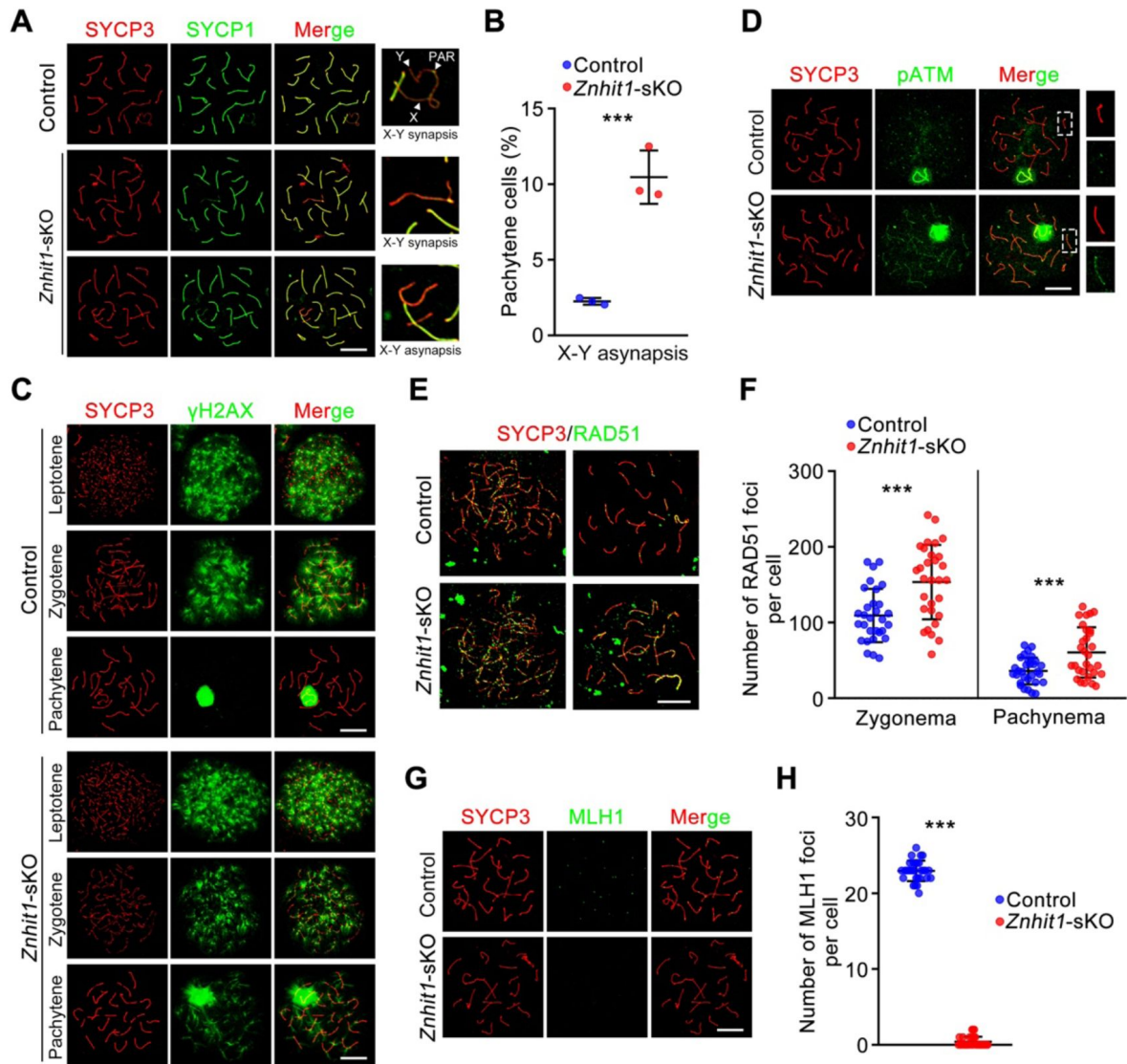


Figure 3

Znhit1 controls meiotic recombination.

(**A** and **B**) Immunostaining of SYCP3 and SYCP1 in spermatocyte chromosome spreads of control or *Znhit1*-sKO mice. X, X chromosome; Y, Y chromosome; PAR, pseudoautosomal region. Quantitative data are shown in (**B**). Scale bar, 10 μ m. (**C**) Immunostaining of SYCP3 and γ H2AX in spermatocyte chromosome spreads of control or *Znhit1*-sKO mice. Scale bar, 10 μ m. (**D**) Immunostaining of SYCP3 and pATM in spermatocyte chromosome spreads of control or *Znhit1*-sKO mice. Scale bar, 10 μ m. (**E** and **F**) Immunostaining of SYCP3 and RAD51 in spermatocyte chromosome spreads of control or *Znhit1*-sKO mice. Quantitative data are shown in (**F**). Scale bar, 10 μ m. (**G** and **H**) Immunostaining of SYCP3 and MLH1 in spermatocyte chromosome spreads of control or *Znhit1*-sKO mice. Quantitative data are shown in (**H**). Scale bar, 10 μ m. All images are representative of $n = 3$ mice per genotype. Data are presented as the mean $\pm$ s.d. *** $p < 0.001$.

Collectively, these results argue that *Znhit1* is essential for multiple meiotic events, including sexual chromosome synapsis, DSB repair, and homologous recombination.

scRNA-seq analysis reveals abnormal pachytene-to-diplotene transition caused by *Znhit1* knockout

We then asked how *Znhit1* regulates spermatocyte development. We isolated testicular cells from control and *Znhit1*-sKO mice at postnatal 16 (P16) and performed scRNA-seq analysis. After quality control, a total of 2,4164 cells (12,212 cells from control, 11,952 cells from *Znhit1*-sKO) were captured, yielding 2,473 genes per cell. As visualized via uniform manifold approximation and projection (UMAP), the single-cell atlas comprised germ cells at various stages and major somatic cell types, including Sertoli cells (*Sox9*⁺), Leydig cells (*Hsd3b1*⁺), peritubular myoid cells (*Acta2*⁺), and macrophages (*Adgre1*⁺) (Figs 4A, 4B and S4A, and Table S2). The scRNA-seq data identified 6 germ cell types based on the expression of specific markers, including undifferentiated spermatogonia (*Zbtb16*⁺), differentiating spermatogonia (*Kit*⁺), preleptotene spermatocytes (*Nacadm*⁺), leptotene spermatocytes (*Meio*⁺), zygotene spermatocytes (*Gml*⁺), and pachytene spermatocytes (*Ldhc*⁺) (Fig 4B and 4C). Examination of germ cell scRNA-seq profiles via unsupervised clustering revealed decreased pachytene spermatocytes after *Znhit1* deletion (Figs 4D, 4E, and S4B). *Pou5f2* is expressed explicitly at the diplotene stage, and we found a lack of *Pou5f2*⁺ diplotene spermatocytes in *Znhit1*-sKO testes (Fig 4F). These results indicate that *Znhit1* is required for male germ cell development beyond the pachytene stage.

Znhit1 deletion impairs meiotic transcriptional activation

Next, we asked whether meiotic transcriptional activation was affected by *Znhit1* deletion. We analyzed changes in gene expression between consecutive spermatocyte stages and identified 4,603 genes that showed significant changes (Fig 5A and Table S3). Of the changed genes between zygotene spermatocytes and pachytene spermatocytes, 1,560 genes were upregulated (referred to as pachytene-activated genes or PGA genes). Moreover, we identified differentially expressed genes (DEGs) in each spermatocyte type between the control and *Znhit1*-sKO groups. The number of DEGs gradually increased from the preleptotene stage to the pachytene stage, with downregulated genes being predominant (Fig 5B and Table S4). In particular, *Znhit1*^{-/-} pachytene spermatocytes exhibited 2,785 downregulated genes, among which were most of the pachytene-activated genes (1,094 out of 1,560, 70.1%; Figs 5B–5D, and S5A). Gene Ontology (GO) enrichment analysis revealed that upregulated genes following *Znhit1* deletion were associated with apoptotic pathways, supporting an observation that *Znhit1* deletion resulted in germ cell clearance through apoptosis. Moreover, downregulated GO terms included germ cell development and DNA recombination (Fig 5E and Table S5).

Interestingly, recent studies showed that cilium organization was essential for spermatocyte development (Mytlis et al., 2022), and *Znhit1* deletion reduced the expression of cilium-related genes in zygotene and pachytene spermatocytes (Fig 5E).

We further conducted bulk RNA-seq experiments with P14 control and *Znhit1*-sKO testicular cells and identified 882 DEGs (812 downregulated genes and 70 upregulated genes) in *Znhit1*-sKO testicular cells (Figs 5F and S6A, and Table S6). Through integrative analysis of bulk and single-cell RNA-seq data, we found that the expression of these 812 genes was activated during the zygotene-to-pachytene transition and *Znhit1* deletion significantly reduced the expression of these genes (Figs 5G, 5H, and S5B). The downregulated genes following *Znhit1* loss were enriched for germ cell-related GO terms, such as cilium formation and spermatid development (Fig S6B). Together, these data argue that *Znhit1* is required for meiotic transcriptional activation.

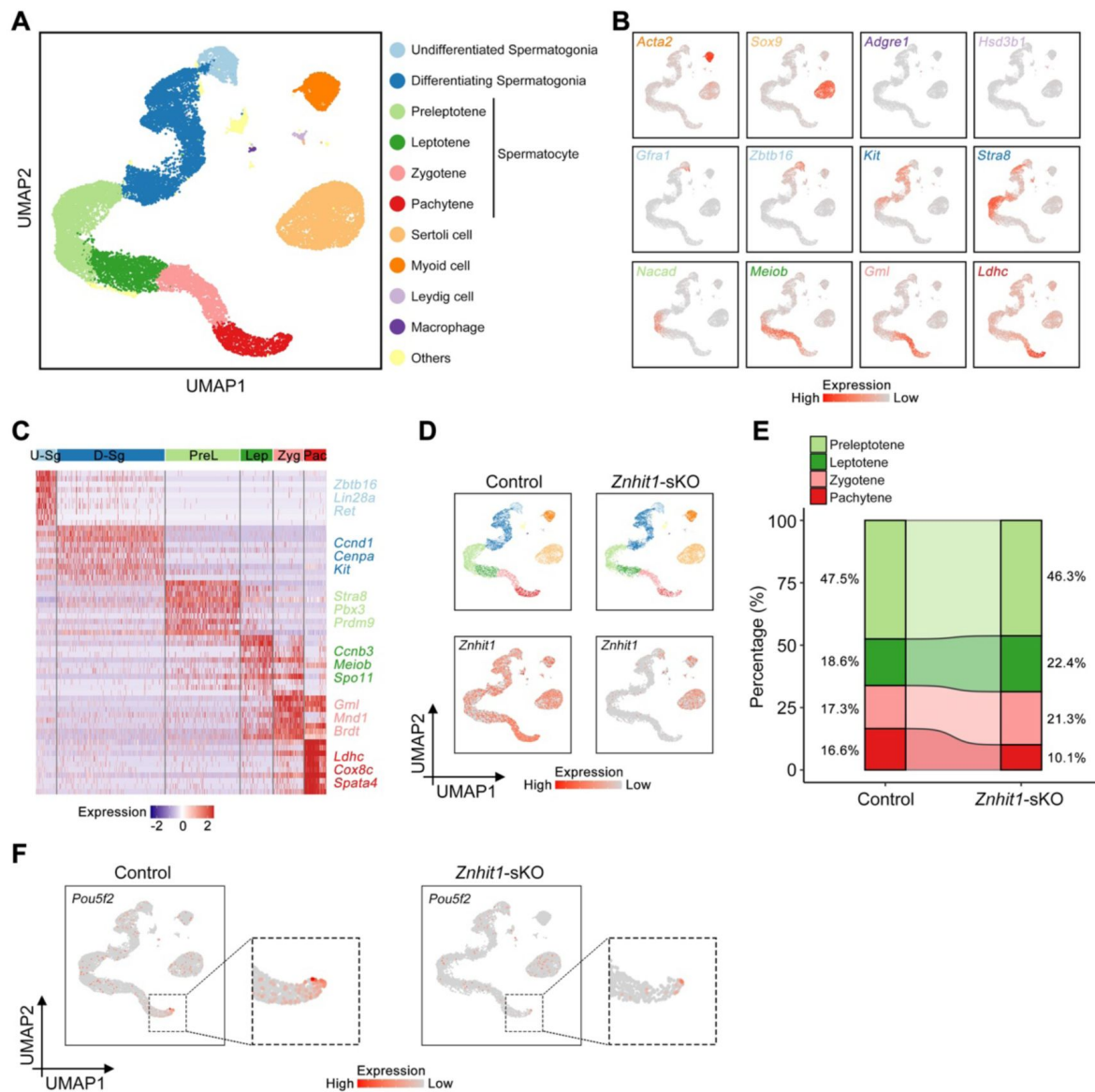


Figure 4

Spermatocyte transcriptome analysis using scRNA-seq.

(A) UMAP plot showing the annotated germ cells captured from both control and *Znhit1*-sKO testicular cells. (B) Expression of selected markers identifying major somatic cell types and germ cell types on the UMAP plot. UMAP plot showing the expression patterns of *Znhit1* in control or *Znhit1*-sKO testicular cells. (C) Heatmap showing the top marker genes of each cell cluster from scRNA-seq data. U-Sg, undifferentiated spermatogonia; D-Sg, differentiating spermatogonia; PreL, preleptotene; Lep, leptotene; Zyg, zygotene; Pac, pachytene. (D) Top: UMAP plot showing the annotated germ cells captured from control or *Znhit1*-sKO testicular cells. Down: UMAP plot showing the expression patterns of *Znhit1* in control or *Znhit1*-sKO testicular cells. (E) Bar plot showing the percentage of major spermatocyte types for control or *Znhit1*-sKO testicular cells. (F) UMAP plot showing the expression patterns of *Pou5f2* in control or *Znhit1*-sKO germ cells.

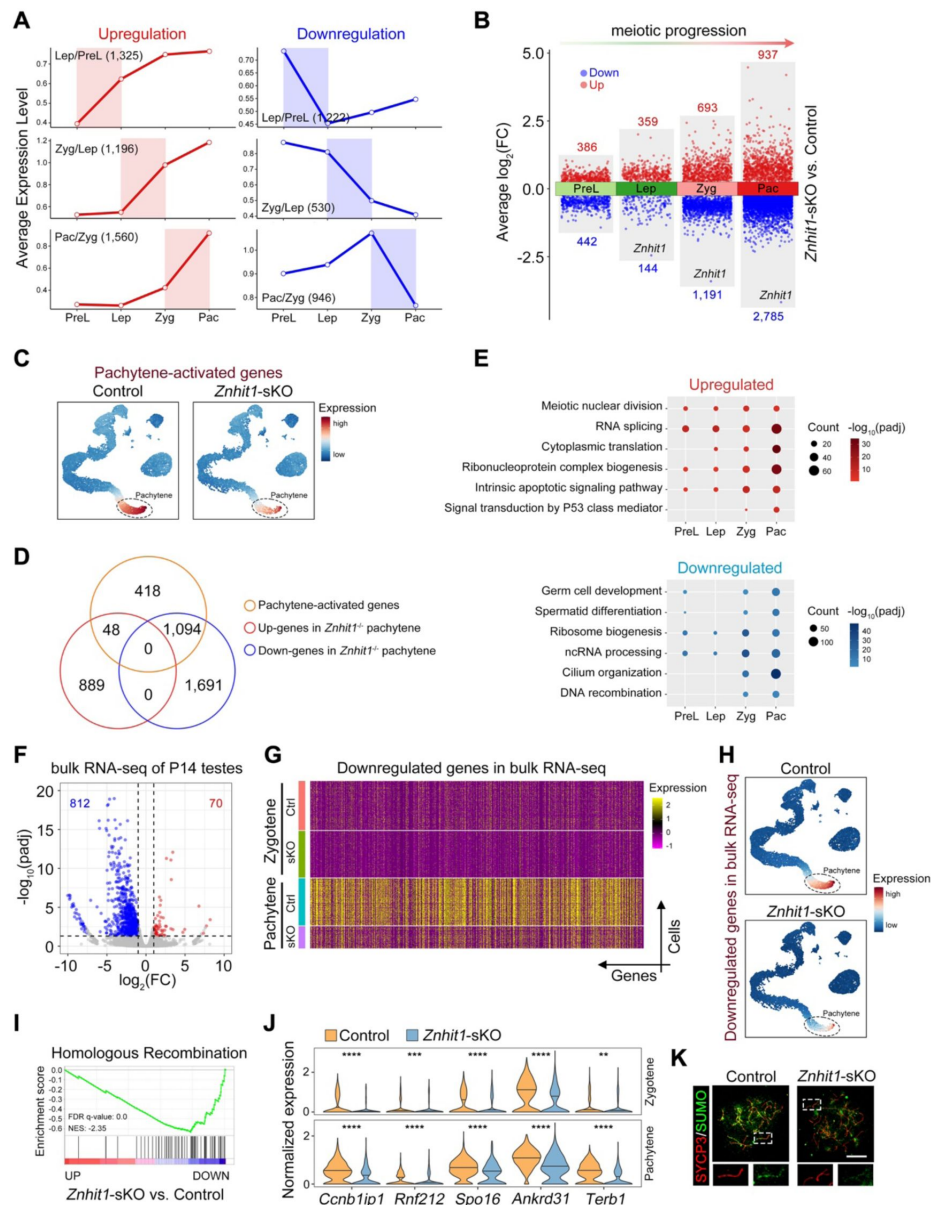


Figure 5

Znhit1 deletion impairs meiotic transcriptional activation.

(A) Line graph showing DEGs (upregulation or downregulation) in two consecutive spermatocyte stages. PreL, preleptotene; Lep, leptotene; Zyg, zygotene; Pac, pachytene. (B) Volcano plot showing the distribution of upregulated and downregulated DEGs for each cell type in spermatocytes between control or *Znhit1*-sKO testicular cells. (C) Cluster annotation on the basis of the expression of pachytene-activated genes in control or *Znhit1*-sKO testicular cells. (D) Venn plots showing the number of shared genes between pachytene-activated genes and DEGs in *Znhit1*-sKO pachytene spermatocytes. (E) Representative shared Gene Ontology (GO) terms of upregulated (upper) and downregulated (lower) DEGs at consecutive spermatocyte stages. (F) Volcano plot showing differentially expressed genes (DEGs) in bulk RNA-seq following *Znhit1* deletion ($n = 2$). (G) Heatmap depicting the expression of the indicated genes at the zygotene and pachytene stages in the scRNA-seq data. (H) Cluster annotation on the basis of the expression of downregulated genes in bulk RNA-seq between control and *Znhit1*-sKO testicular cells. (I) GSEA of RNA-seq data for the control and *Znhit1*-sKO testicular cells. Selected gene sets encoded products related to homologous recombination. (J) Violin plots showing downregulated DEGs of *Znhit1*-deficient zygotene cells and pachytene cells related to homologous recombination. (K) Immunostaining of SYCP3 and SUMO2/3 in spermatocyte chromosome spreads of control or *Znhit1*-sKO mice. Scale bar, 10 μ m. cluster genes. ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

Consistent with meiotic recombination defects in *Znhit1* mutants, gene set enrichment analysis (GSEA) revealed that the expression of HR-related genes was globally downregulated following *Znhit1* loss (**Fig 5I** and Table S7). The top genes associated with this function included *Ccnb1ip1*, *Rnf212*, *Spo16*, *Ankrd31*, and *Terb1* (Boekhout et al., 2019; Papanikos et al., 2019; Qiao et al., 2014; Rao et al., 2017; Reynolds et al., 2013; Shibuya et al., 2014; Wang et al., 2019; Zhang et al., 2019) (**Figs 5J**, S6C, and S6D). As expected, the Rnf212-dependent sumoylation level decreased in *Znhit1*^{-/-} spermatocytes (**Fig 5K**). These results suggest that *Znhit1* regulates meiotic recombination by activating HR-related genes.

***Znhit1* deletion impairs H2A.Z incorporation into pachytene chromatin**

We then asked how *Znhit1* regulates transcription. *Znhit1* is a subunit of the SRCAP complex that facilitates the incorporation of the histone variant H2A.Z by replacing H2A (Fig S7A). Immunostaining against H2A.Z showed a gradual increase in H2A.Z accumulation on autosomes from the leptotene to the diakinesis stage, while the H2A.Z signal was markedly low on the X-Y body (**Fig 6A**). *Znhit1* loss decreased H2A.Z staining on the pachytene cells, suggesting that *Znhit1* is required for maintaining H2A.Z integrity during meiosis.

To probe the H2A.Z targets, we performed chromatin immunoprecipitation followed by sequencing (ChIP-seq) against H2A.Z in P14 testicular cells. Using the ChromHMM model to annotate H2A.Z signals with meiotic chromatin states defined by published epigenetic markers (Spruce et al., 2020), we found that H2A.Z sites were primarily enriched in chromatin state 1 and state 2 (promoters and enhancers) but were less enriched in recombination hotspots (**Fig 6B** and **6C**), consistent with previously published results (Spruce et al., 2020). Quantitative analysis showed significant H2A.Z binding reduction at 37.9% (23,926 of 63,109) of H2A.Z-bound genome sites following *Znhit1* deletion (**Fig 6D**). Comparison analysis with bulk RNA-seq data identified 480 downregulated DEGs with decreased H2A.Z signals (**Fig 6E**). ~41.3 and 37.8% of the downregulated H2A.Z peaks occupied promoters and enhancers, respectively. At promoters, these downregulated H2A.Z preferentially occupied active promoters (H3K4me3 only), while H2A.Z occupied three types of enhancers: active, poised, and primed enhancers (**Figs 6F**, S7B, and S7C). These findings indicate that *Znhit1* deletion impairs H2A.Z incorporation into pachytene chromatin.

H2A.Z deposition cooperates with A-MYB to regulate transcription

We further asked how *Znhit1*/H2A.Z deposition facilitates lineage-specific gene expression. We performed TF motif analysis using active promoters/enhancers with decreased H2A.Z signals. This analysis identified several MYB family TFs, including MYB, A-MYB, and B-MYB (**Fig 7A**). Transcriptional network analysis conducted by SCENIC further identified A-MYB (encoded by *Mybl1* gene) as the core regulatory TF in spermatocytes (**Fig 7B**). We reanalyzed RNA-seq data in P14 testes from A-MYB-deficient mice (Li et al., 2013) and found that A-MYB-deficient DEGs correlated with those observed in *Znhit1*-deficient testicular cells (**Fig 7C**). Although *Znhit1* deletion didn't affect *Mybl1* mRNA expression, transcriptome analyses showed that the expression of A-MYB target genes was significantly reduced in *Znhit1*^{-/-} pachytene spermatocytes (**Figs 7D**, **7E**, S8A, and S8B). These results suggest that *Znhit1*^{-/-} and *Mybl1*^{-/-} pachytene spermatocytes share common gene expression alterations.

Next, we compared the genome binding sites of H2A.Z and A-MYB. Utilizing published A-MYB ChIP-seq data from P14 testicular cells (Li et al., 2013), we identified 6,088 A-MYB binding signals, with 78.1% (4755) of these coinciding with H2A.Z peaks (**Fig 7F**). Annotation of genomic features showed that approximately 66.52% and 12.02% of these overlapping sites occupied active promoters and enhancers, respectively, representing a stronger enrichment compared to other regulatory elements (**Fig 7G**). Genes associated with H2A.Z and A-MYB binding included

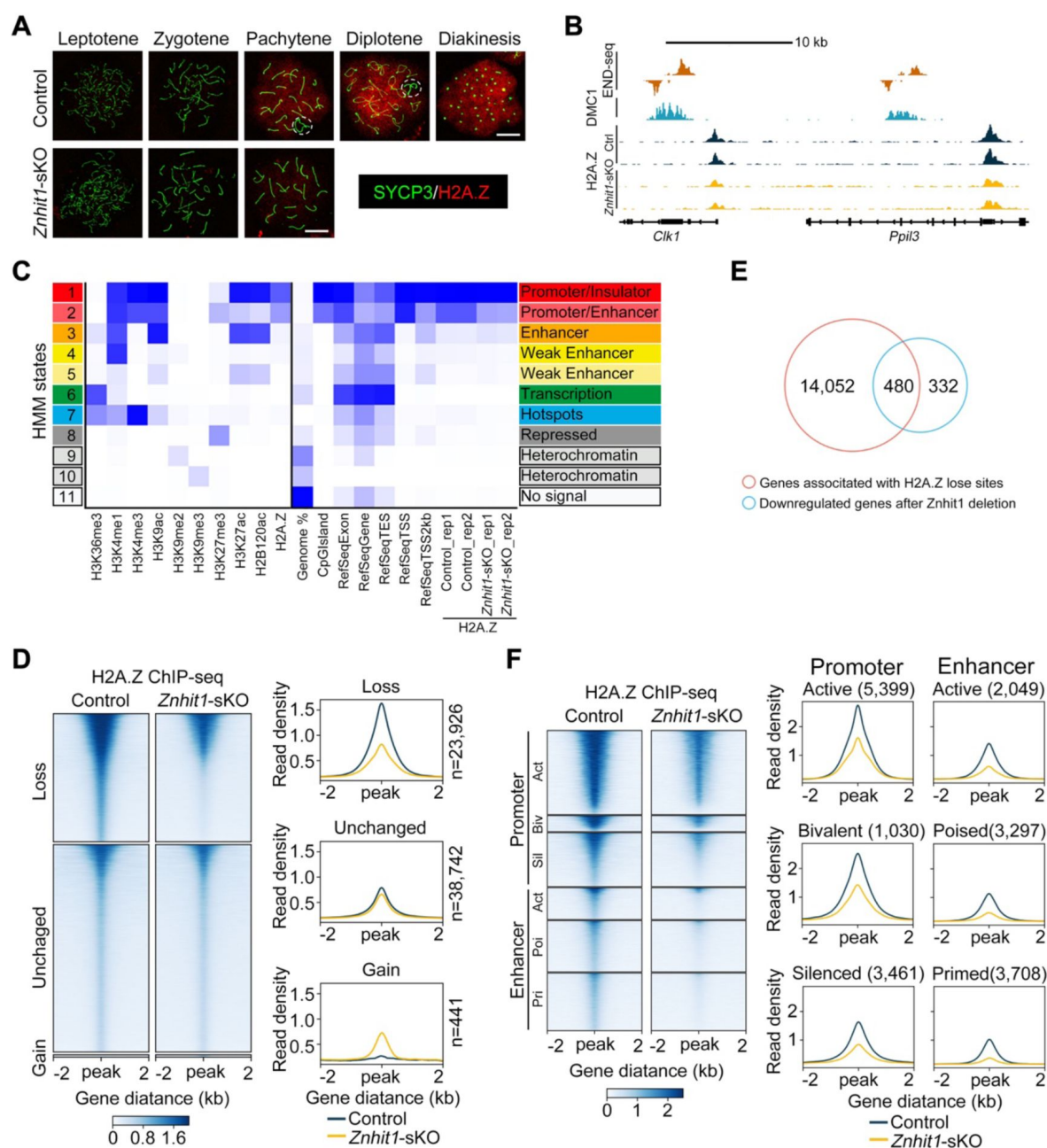


Figure 6

Znhit1 regulates H2A.Z binding on promoter and enhancer regions.

(A) Immunostaining of SYCP3 and H2A.Z in spermatocyte chromosome spreads of control or *Znhit1*-sKO mice. The dashed circle indicates X-Y chromosomes. Scale bar, 10 μ m. (B) Representative tracks of END-seq, DMC1-SSDS, and H2A.Z ChIP-seq in wild-type, control, or *Znhit1*-sKO testicular cells. (C) Heatmaps of chromatin states produced by ChromHMM and showing different enrichment for H2A.Z in control or *Znhit1*-sKO testicular cells. (D) Heatmaps and mean plots of H2A.Z ChIP-seq signal in control or *Znhit1*-sKO testicular cells ($n = 2$). (E) Venn diagram showing the overlap between decreased H2A.Z-bound genes and downregulated genes in *Znhit1*-sKO testicular cells. (F) Heatmaps and mean plots of downregulated H2A.Z ChIP-seq signals surrounding different types of promoters and enhancers.

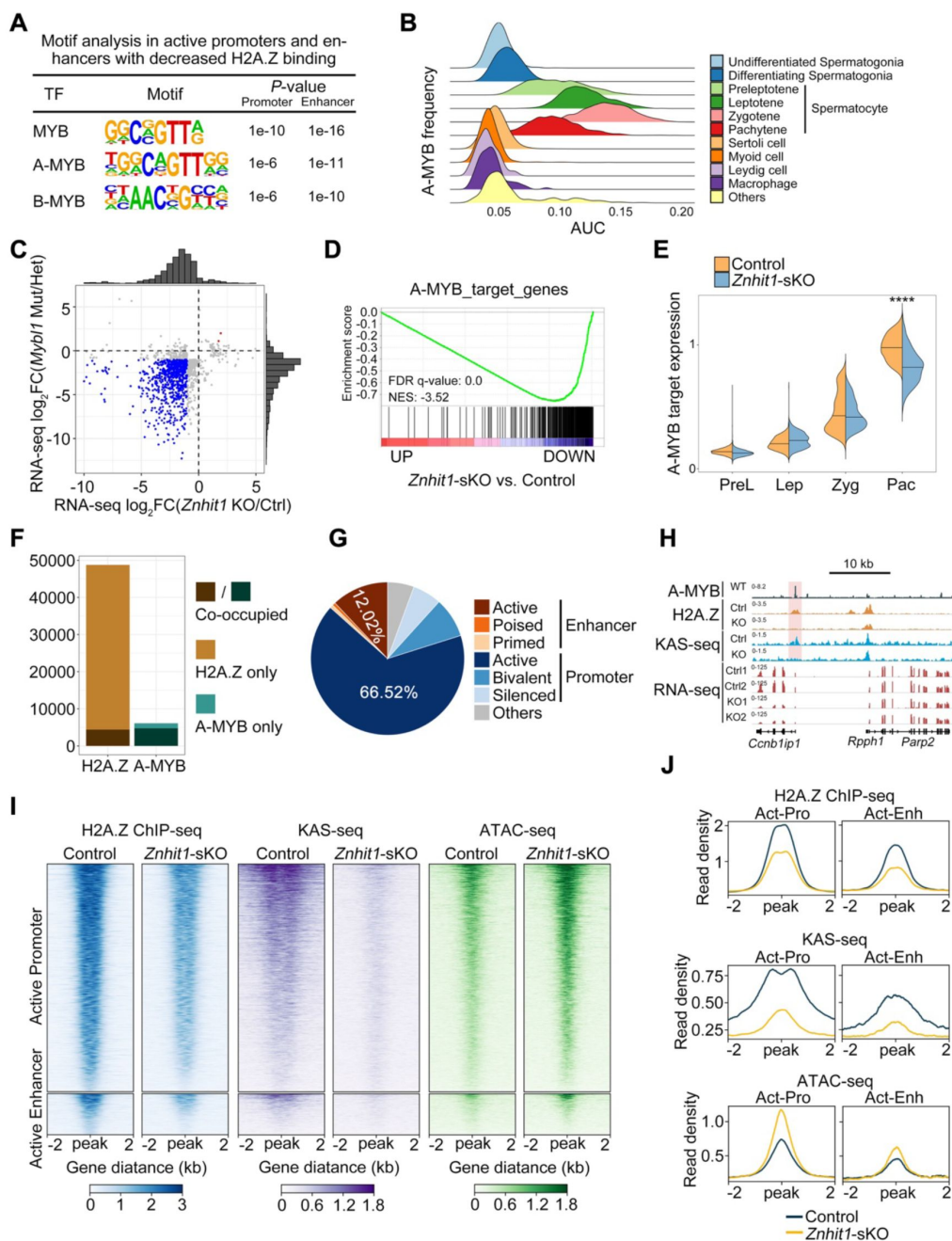


Figure 7

***Znhit1*/H2A.Z/A-MYB axis regulates enhancer activation and gene expression.**

(A) Motif analysis of active promoters or active enhancers with decreased H2A.Z binding for putative transcription factor (TF)-binding sites using the HOMER database. (B) Ridge map showing the A-MYB activity (AUC) at each cell type. (C) Scatter plot with marginal histograms comparing the fold change of DEGs from A-MYB-related RNA-seq data with *Znhit1*-related RNA-seq data in P14 testicular cells. (D) GSEA of RNA-seq data for the control and *Znhit1*-sKO testicular cells. Selected gene sets encoded products related to A-MYB target genes. (E) Violin plots showing the expression of A-MYB target genes at consecutive spermatocyte stages. (F) Histogram showing H2A.Z and A-MYB binding sites. (G) Pie chart showing the distribution of H2A.Z and A-MYB co-binding sites. (H) Representative H2A.Z ChIP-seq tracks, KAS-seq signals, and RNA-seq signals of the *Ccnb1ip1* locus in the control and *Znhit1*-sKO testicular cells, compared with A-MYB ChIP-seq peaks. (I and J) Heatmaps and mean plots of H2A.Z ChIP-seq, KAS-seq, and ATAC-seq showing signal changes within A-MYB and H2A.Z co-binding active promoters or active enhancers. **** $p < 0.0001$.

essential HR-related genes, such as *Ccnb1ip1* (encode the Hei10 protein) (Fig 7H). Moreover, we showed that *Znhit1* deletion resulted in a marked reduction in H2A.Z signals and nascent transcription identified by KAS-seq signals (Fig 7I and 7J).

Previous studies have shown the fundamental role of A-MYB in activating meiotic enhancers (Maezawa et al., 2020). Enhancer activity is linked to an increased formation of enhancer single-stranded DNA (ssDNA), which can be evaluated using KAS-seq (Wu et al., 2020). We found that the removal of *Znhit1* resulted in abolished ssDNA at active enhancers (Fig 7I and 7J). Interestingly, the chromatin accessibility detected by ATAC-seq at active enhancers increased after *Znhit1* deletion. These findings suggest that *Znhit1*/H2A.Z deposition is essential for enhancer activation.

Together, these results demonstrate the critical role of the *Znhit1*/H2A.Z/A-MYB axis in governing transcriptional activation and enhancer activity.

Discussion

Meiosis plays a crucial role in the production of haploid gametes. Any disturbances during meiotic progress are known to lead to infertility and congenital diseases. Here we delineate the multifaceted functions of the chromatin remodeler *Znhit1* in regulating meiotic progression. Our study provides convincing data to show that the *Znhit1*/H2A.Z/A-MYB axis controls spermatocyte development, meiotic recombination, and PGA. Thus, these findings underscore the significance of specific chromatin structures in governing appropriate transcriptional reprogramming and precise meiotic recombination.

The role of H2A.Z in DNA injury and repair has been extensively studied (Colino-Sanguino et al., 2022; Dong et al., 2014; Xu et al., 2012; Yamada et al., 2018), but its function in programmed DSB formation and repair during mammalian meiosis remains unknown. To address this question, we generated *Znhit1* conditional knockout mice specifically during male meiosis to examine the role of *Znhit1*-mediated H2A.Z deposition in the formation and repair of programmed DSB. One new observation in this study is that *Znhit1* loss is dispensable for DSB formation but necessary for DNA break repair during meiosis. Intriguingly, unlike the observations in plant meiosis, we found that H2A.Z does not directly bind DNA recombination sites in mammalian spermatocytes (Choi et al., 2013; Spruce et al., 2020; Wang and Copenhagen, 2018). Moreover, we showed that *Znhit1* deletion delays recombinational repair, but has limited impacts on DNA resection, ultimately causing defective meiotic crossover formation. Importantly, our study demonstrates that *Znhit1* is essential for the expression of meiotic recombination-related genes, such as *Ccnb1ip1* and *Rnf212*, implying that *Znhit1*/H2A.Z modulates meiotic recombination through transcriptional regulation. Therefore, our data reveals a different mechanism underlying meiotic recombination between plants and mammals.

Pachytene spermatocytes utilize a highly specific mechanism of meiotic surveillance, the pachytene checkpoint, to prevent aneuploidy formation by removing abnormal germ cells with incomplete chromosome synapsis and defective homologous recombination (Huang and Roig, 2023; Roeder and Bailis, 2000; Subramanian and Hochwagen, 2014). Two checkpoint pathways exist in male meiosis: one responding to unrepaired DSBs and the other triggered by MSCI failure (Royo et al., 2013). However, whether epigenetic regulation is involved in meiotic surveillance is not well defined. Previous studies have reported that upon stimulation by unrepaired DSBs, the MRE11-ATM-CHEK2 pathway is activated to eliminate aberrant germ cells (Marcet-Ortega et al., 2017). Our study found that *Znhit1* deletion impairs the removal of ATM phosphorylation signals on autosomes in pachynema, ultimately causing meiotic pachytene arrest and apoptosis. Therefore, these findings demonstrate that *Znhit1* acts as an important chromatin factor involved in the regulation of the pachytene checkpoint.

It has long been noticed that in meiotic prophase I, chromatin that has not yet completed chromosomal synapses is transcriptionally inactive, known as meiotic silencing of unsynapsed chromatin (MSUC) (Turner et al., 2005 [\[1\]](#)). When germ cells enter the pachytene stage, a large number of protein-coding genes and non-coding RNAs begin to be actively expressed, known as PGA. Previous studies have shown that transcription factors A-MYB and BRDT are involved in transcriptional activation during meiotic prophase, but how PGA takes place in the chromatin context is unclear. In this study, we found that the expression of the chromatin remodeler *Znhit1* is specifically upregulated during the zygotene-to-pachytene transition. We also observed that H2A.Z deposition is enriched in the autosomes but less in the sex chromosome at the pachytene stage. *Znhit1* deletion in spermatocytes repressed pachytene gene activation globally and reduced chromosome-wide H2A.Z deposition, supporting a central role of *Znhit1* in PGA regulation and chromatin remodeling. It has been known that A-MYB is a transcription factor that controls pachytene transcription activation. We found that *Znhit1*-deficient spermatocytes phenocopied abnormal meiotic phenotypes in A-MYB mutants, such as X-Y synapsis failure, impaired DSB repair, and defective HR (Alexander et al., 2023 [\[2\]](#); Bolcun-Filas et al., 2011 [\[3\]](#); Li et al., 2013 [\[4\]](#); Maezawa et al., 2020 [\[5\]](#)). Our results also revealed that *Znhit1*/H2A.Z cooperates with A-MYB to regulate PGA gene activation. Therefore, our study illuminates the molecular mechanisms underlying the fundamental question of how PGA is regulated.

The histone variant H2A.Z is enriched at gene promoters and regulatory regions, but there is ongoing debate regarding its involvement in transcriptional regulation. A recent paper reported that H2A.Z knockout in post-mitotic muscle cells has limited effects on gene expression (Belotti et al., 2020 [\[6\]](#)). In this study, we utilized the meiotic prophase as a model system (where DNA replication does not occur) to study H2A.Z's function in transcriptional regulation. We found that *Znhit1*-mediated H2A.Z deposition, independent of DNA replication, is indispensable for the transcriptional activation of a large number of meiotic genes. One explanation of this conflicting phenomenon is that H2A.Z dynamics but not stable H2A.Z accumulation is essential for priming transcriptional changes.

Through the integration of functional and molecular evidence, our findings establish the critical involvement of *Znhit1*-dependent chromatin remodeling in the orchestration of meiotic progression and coordination of various meiotic processes, such as HR and PGA. Furthermore, we pinpoint the pivotal role played by the *Znhit1*/H2A.Z/A-MYB axis in driving transcriptional reprogramming during meiotic prophase. Taken together, this study deepens our understanding of the interplay between epigenetic regulation and mammalian meiosis.

Materials and Methods

Animals

Znhit1^{fl/fl} mice have been previously described (Zhao et al., 2019 [\[7\]](#)) and are available from the Model Animal Research Center of Nanjing University (MARC, Nanjing, China). The *Stra8-cre* knock-in mouse line was kindly provided by Dr. Ming-Han Tong (Lin et al., 2017 [\[8\]](#)). All mice were maintained on the C57BL/6J background.

Germ cell-specific *Znhit1* knockout mice (*Znhit1*^{fl/fl}; *Stra8-cre*) were obtained by crossing *Znhit1*^{fl/+}; *Stra8-cre* mice with *Znhit1*^{fl/fl} mice. All mice were housed in the SPF (Specific-Pathogen-Free) animal facility with standard 12 h light/dark cycles and standard temperature (22 to 24°C). All mice were provided with ad libitum access to standard laboratory food and water. All experiments in this study were performed following relevant guidelines and approved by the Animal Care and Use Committee of Fudan University.

Histological and immunohistochemical analysis

Testes were fixed in modified Davidson's fixative as previously described (Latendresse et al., 2002), embedded in paraffin, and sectioned. For periodic acid-Schiff (PAS)-hematoxylin staining, 5 μ m testis sections were deparaffinized, rehydrated, and stained with Schiff's reagent and hematoxylin solution. For immunofluorescent staining, 5 μ m testis sections were retrieved by sodium citrate antigen retrieval buffer (pH 6.0) and blocked with 5% BSA in PBS for 30 min at room temperature. The sections were later incubated overnight at 4°C with primary antibodies as follows: mouse anti-SYCP3 (Abcam, ab97672), rabbit anti-HSPA2 (Abcam, ab108416), rabbit anti-pH3 (Millipore, H0412), rabbit anti-H1T (Invitrogen, PA5-51200), or lectin PNA (Sigma-Aldrich, L7381). On the following day, secondary fluorescein-conjugated antibodies and DAPI (Sigma-Aldrich, D9542) were added for 1 h, followed by Fluoromount-G mounting (Southern Biotech, 0100-01).

Images were analyzed using the confocal microscope.

TUNEL staining was carried out using the DeadEnd™ Fluorometric TUNEL System (Promega, G3250) according to the manufacturer's instructions.

Immunostaining of spermatocyte chromosome spreads

Spermatocyte chromosome spreads were prepared as previously described (Alavattam et al., 2018). Briefly, the clumps of seminiferous tubules were transferred to the hypotonic extraction buffer (30 mM Tris base, 17 mM trisodium citrate, 5 mM EDTA, 50 mM sucrose, 5 mM dithiothreitol (DTT), and 1× protease inhibitor, pH 8.2) and incubated on ice for 1.5 h. The clumps of tubules were then transferred to 30 μ L of ice-cold 100 mM sucrose and mashed gently using tweezers to obtain a cell suspension. An additional 30 μ L of ice-cold 100 mM sucrose was added to the cell suspension and mixed several times. Positively charged slides were incubated in the ice-cold fixation solution (2% paraformaldehyde, 0.1% Triton X-100, and 0.02% SDS, pH 9.2) for 3 min. 30 μ L of the diluted cell suspension was applied to the slides and incubated in humid chambers at room temperature for 2 h. The slides were washed with 0.4% Photo-Flo 200 and stored at -80°C.

For immunostaining of spermatocyte chromosome spreads, slides were washed with PBS and blocked with 5% BSA in PBS for 30 min at room temperature. The slides were stained with primary antibodies as follows: rabbit anti-SYCP3 (Abcam, ab15093), mouse anti-SYCP3 (Abcam, ab97672), rabbit anti-SYCP1 (Abcam, ab15090), mouse anti- γ H2AX (BioLegend, 613401), rabbit anti-pATM (Millipore, 05-740), rabbit anti-RAD51 (Proteintech), rabbit anti-RPA2 (Proteintech, 10412-1-AP), rabbit anti-MLH1 (Cell Signaling Technology, 3515T), rabbit anti-H2A.Z (Abcam, ab4174).

Znhit1 in situ hybridization

Testis sections (5 μ M) at the indicated times were prepared for *Znhit1* in situ hybridization with the RNAscope kit (Advanced Cell Diagnostics, 323100) according to the manufacturer's instructions.

Quantitative reverse transcription PCR (RT-qPCR)

Total RNA was isolated using the RNeasy Mini-plus Kit (Qiagen, 74134) and then reverse-transcribed into complementary DNA (cDNA) with the GoScript Reverse Transcription System (Promega, A5003). cDNA was used as the template for the quantitative PCR assay using 2×SYBR Green qPCR Master Mix (Bimake, B21202). Quantitative PCR primers are listed in Table S8.

RNA-seq library generation and sequencing

Total RNA from fresh testes was isolated and mRNA was purified with magnetic beads (Vazyme, N401-01). Then, mRNA was fragmented and processed to generate RNA-seq libraries (Vazyme, NR605-01). Over 40 million reads were obtained per sample using the Illumina NovaSeq platform for 2 independent biological replicates.

Single-cell RNA sequencing (scRNA-seq)

P16 control and *Znht1*-sKO testes were digested by collagenase IV and trypsin at 37°C for 10 min to obtain testicular cell suspensions. scRNA-seq libraries were constructed using a 10 × Genomics kit and sequenced on the Illumina platform.

ChIP-seq library generation and sequencing

ChIP-seq library generation was performed as previously described. Briefly, P14 control and *Znht1*-sKO testes were digested by collagenase IV and trypsin at 37°C for 10 min, crosslinked with 1% formaldehyde for 10 min, and quenched with glycine. The cells were lysed and sheared with a Bioruptor Plus machine for 20 min. Then, 2 µg of anti-H2A.Z antibody (Abcam, ab4174) was added to the sonicated chromatin and incubated overnight at 4°C. The following day, 20 µL of protein G beads were added and incubated for 2 h at 4°C. The beads were washed and reverse-crosslinked for 4 h at 65°C. DNA was extracted using phenol-chloroform and subjected to library construction using the VAHTS Universal DNA Library Prep Kit for Illumina (Vazyme, ND607-01) according to the manufacturer's instructions. Over 30 million reads were obtained per sample using the Illumina NovaSeq platform for 2 independent biological replicates.

KAS-seq library and ATAC-seq library generation and sequencing

Testes were digested by collagenase IV and trypsin at 37°C for 10 min. Pachytene cells were sorted using fluorescence-activated cell sorting (FACS) as previously described (Long et al., 2017 [DOI](#)). For KAS-seq, pachytene cells were labeled with N3-kethoxal, and DNA was isolated using the DNA Clean and Concentrator kit (Zymo, D4013) and subjected to library construction using Q5 high-fidelity DNA polymerase (New England Biolabs, M0544S). Genomic DNA was fragmented using Tn5 transposase and subjected to library construction. For ATAC-seq, pachytene nuclei were isolated and fragmented with Tn5 transposase. DNA was isolated using the DNA Clean and Concentrator kit (Zymo, D4013) and subjected to library construction using Q5 high-fidelity DNA polymerase (New England Biolabs, M0544S). For high-throughput sequencing, over 30 million reads were obtained per sample using the Illumina NovaSeq platform for 2 independent biological replicates.

RNA-seq data analysis

We performed RNA-seq analysis as described previously (Pertea et al., 2016 [DOI](#)). Briefly, after removing adapters using Cutadapt (v2.5) (Kechin et al., 2017 [DOI](#)), paired-end reads were aligned to the annotated mouse transcripts (mm10 Gencode vM23 release) using Hisat2 (v2.2.1) (Kim et al., 2015 [DOI](#); Kim et al., 2019 [DOI](#)). Gene expression levels were calculated using StringTie (v2.2.1) (Pertea et al., 2015 [DOI](#)). Read counts were calculated using a Python script (prepDE.py3) provided by the StringTie development team. Differentially expressed genes were identified using the R package DESeq2 (v1.38.3) (Love et al., 2014 [DOI](#)). Genes with read counts > 50 in at least one sample were kept for further analysis. A given gene was considered to significantly change if the adjusted P value (padj) was < 0.05, the P value was < 0.01, and the fold-change was ≥ 2.

Gene Set Enrichment Analysis (GSEA) was carried out using GSEA software (Subramanian et al., 2005 [DOI](#)). GO analysis was performed using clusterProfiler (v4.6.2) (Wu et al., 2021 [DOI](#)).

scRNA-seq data analysis

FASTQ files were run through CellRanger (v7.1.0) software with default parameters for de-multiplexing, aligning reads with STAR software to mm10, and counting unique molecular identifiers (UMIs). As input files of the Seurat R package (v4.4.0), the filtered gene expression matrices were then used for downstream analyses (Butler et al., 2018 [\[1\]](#)). Low-quality cells were filtered (expressing < 500 or >6,000 unique gene counts and >15% mitochondrial reads). Principal component analysis was performed on SCT-transformed data using 3,000 variable genes. The top 50 principal components were used for clustering and visualized using the UMAP algorithm in the Seurat R package. The “FindAllMarkers” function of the Seurat R package was used to calculate cluster-specific genes. Marker genes for each cluster are shown in Table S2.

The “FindMarkers” function of the Seurat R package was used to identify differentially expressed genes (DEGs) for spermatocytes (preleptotene, leptotene, zygotene, and pachytene). Only those with $|\text{avg_logFC}| > 0.25$ and $\text{p_val_adj} < 0.05$ were considered as DEGs. For the transcriptional regulatory network analysis, the raw count matrix was used by the pyscenic (v0.12.1) workflow using default parameters (Aibar et al., 2017 [\[2\]](#)). Then, the output loom file was opened by the SCENIC R package (v1.3.1), and AUC values of regulons were extracted for visualization of downstream transcription factor activity.

ChIP-seq and KAS-seq data analysis

For analyzing ChIP-seq and KAS-seq data, we used the ENCODE ChIP-seq pipeline (v2.2.1, <https://github.com/ENCODE-DCC/chip-seq-pipeline2> [\[3\]](#)). Raw reads were cleaned by Cutadapt (v2.5) (Kechin et al., 2017 [\[4\]](#)). After removing adapters, clean reads were aligned against the mouse mm10 genome using bwa (v0.7.17) (Li and Durbin, 2009 [\[5\]](#)). Then, sam files were converted to bam files using samtools (v1.9) (Li et al., 2009 [\[6\]](#)). PCR duplicates were removed using Picard (v2.20.7) (<https://broadinstitute.github.io/picard/> [\[7\]](#)). Histone ChIP-seq peaks (such as H2A.Z and histone marks) and KAS-seq peaks were called using MACS2 (v2.2.4) (Zhang et al., 2008 [\[8\]](#)), while transcription factor ChIP-seq peaks (such as A-Myb) were called using the R packages spp (v1.15.5) (<https://github.com/hms-dbmi/spp> [\[9\]](#)). The conservative narrow peaks were used for downstream analysis.

For visualization, bam files were converted to bigWig files using deepTools (v3.3.1) (Ramirez et al., 2014 [\[10\]](#)), and bigWig files were used to calculate tag density under 50 bp resolution. In addition, CPM was used to normalize the number of reads per bin. We used the R package DiffBind (v3.8.4) (<http://bioconductor.org/packages/release/bioc/html/DiffBind.html> [\[11\]](#)) for differential peak analysis and the R package rGREAT (v2.1.11) (Gu and Hubschmann, 2023 [\[12\]](#)) for peak annotation. We used ComputeMatrix and plotHeatmap of deepTools to generate count matrices and heatmaps, respectively. H2A.Z enrichment within chromatin states was annotated using the ChromHMM model (Ernst and Kellis, 2012 [\[13\]](#); Spruce et al., 2020 [\[14\]](#)). TF motif enrichment was calculated using HOMER (v4.11) (Heinz et al., 2010 [\[15\]](#)). We used deepTools to draw Pearson’s correlation scatter plots for the correlation between replicates of each experiment.

For promoter and enhancer type analysis, H3K27ac ChIP-seq data in wild-type pachytene spermatocytes were downloaded from the published data (GSE107398) (Adams et al., 2018 [\[16\]](#)); ChIP-seq data for H3K4me1, H3K4me3, and H3K27me3 in wild-type pachytene spermatocytes were downloaded from the published data (GSE132446) (Chen et al., 2020 [\[17\]](#)).

ATAC-seq data analysis

ATAC-seq data were analyzed using a standardized ENCODE ATAC-seq pipeline (v2.2.2, <https://github.com/ENCODE-DCC/atac-seq-pipeline> [\[18\]](#)). The mouse reference genome (mm10) was used in the pipeline. Trimming, mapping, and duplicate-removing were performed as the ENCODE pipeline

suggested using Cutadapt (v2.5) (Kechin et al., 2017 [DOI](#)), Bowtie2 (v2.3.4.3) (Langmead et al., 2009 [DOI](#)), and Picard (v2.20.7) (<https://broadinstitute.github.io/picard/> [DOI](#)) respectively. MACS2 (v2.2.4) (Zhang et al., 2008 [DOI](#)) was used for peak calling.

Statistics and reproducibility

Statistical analyses were performed using R and Prism 9. Data are presented as means $\pm$ s.d. unless otherwise indicated. The two-tailed, unpaired, Student's t-test, Mann–Whitney test, or one-way or two-way ANOVA were performed to analyze statistical significance.

Supplementary Materials

This manuscript file includes Figs. S1 to S9 and Tables S1 and S8.

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Author Contributions

S.S. and X.L. conceived and designed the study; S.S. performed most of the experiments with the help of Y.J., Q.Z., H.P., F.H., X.Z., Y.G., X.Y., K.G., W.W., H.L., Z.S., Y.S., X.T., M.Y., and R.L.; Y.J. performed the bioinformatics analysis; X.L. supervised the work and acquired the funding support; and S.S. and X.L. wrote the manuscript, with contributions from all authors.

Competing interests

The authors declare that they have no competing interests.

Data and materials availability

The NGS data generated in this study were deposited to the NCBI SRA database under accession numbers SRP467214 (RNA-seq and ChIP-seq data) and SRP467448 (ATAC-seq and KAS-seq data).

Supplementary Materials

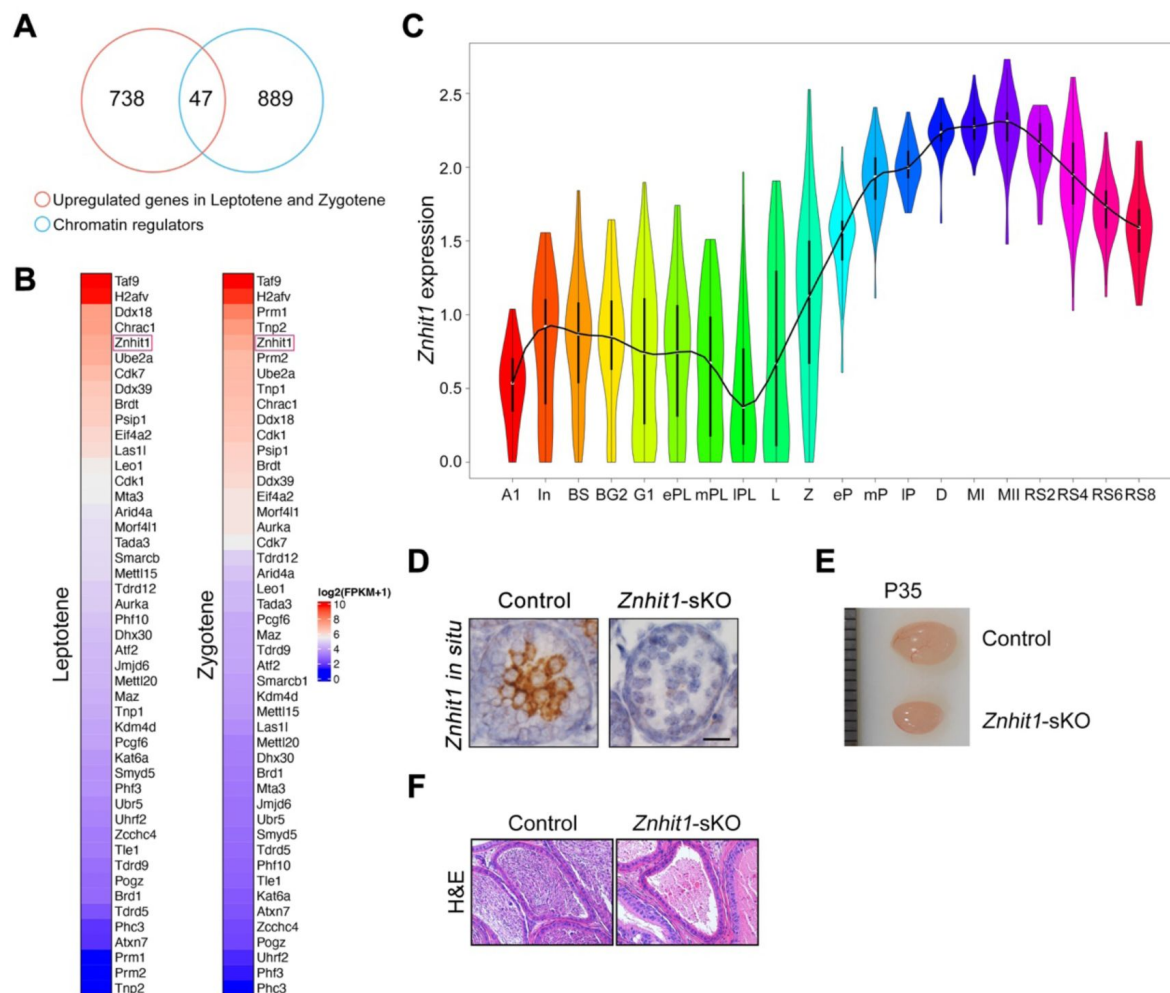


Figure S1.

Znhit1 expression during meiotic progression and spermatocyte-specific deletion of Znhit1 with Stra8-cre.

(A and B) Venn diagram and heatmaps showing chromatin regulators highly expressed during leptotene and zygotene spermatocytes. (C) Violin plot showing Znhit1 transcription level during spermatogenesis. Gene expression profiles derived from published data (Chen et al, 2018 [DOI](#)). (D) Znhit1 in situ hybridization in P14 testis sections. (E) Testis size comparison between control and Znhit1-sKO mice at P35. (F) Hematoxylin and eosin (H&E) - stained testis sections from control and Znhit1-sKO mice at P60.

Figure S2.

H1T⁺ pachytene spermatocytes are detected in *Znhit1*-sKO testes.

(A) Immunostaining of H1T in testis sections from control and *Znhit1*-sKO mice at P35. Scale bar, 20 μ m.

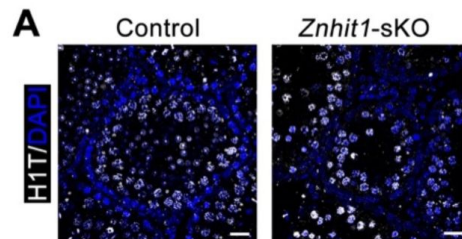
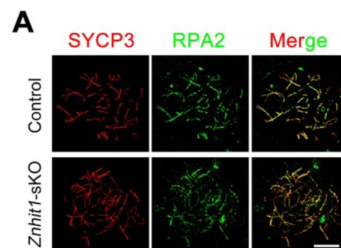


Figure S3.

***Znhit1* controls meiotic recombination.**

(A) Immunostaining of SYCP3 and RPA2 in spermatocyte chromosome spreads of control or *Znhit1*-sKO mice. Scale bar, 10 μ m.



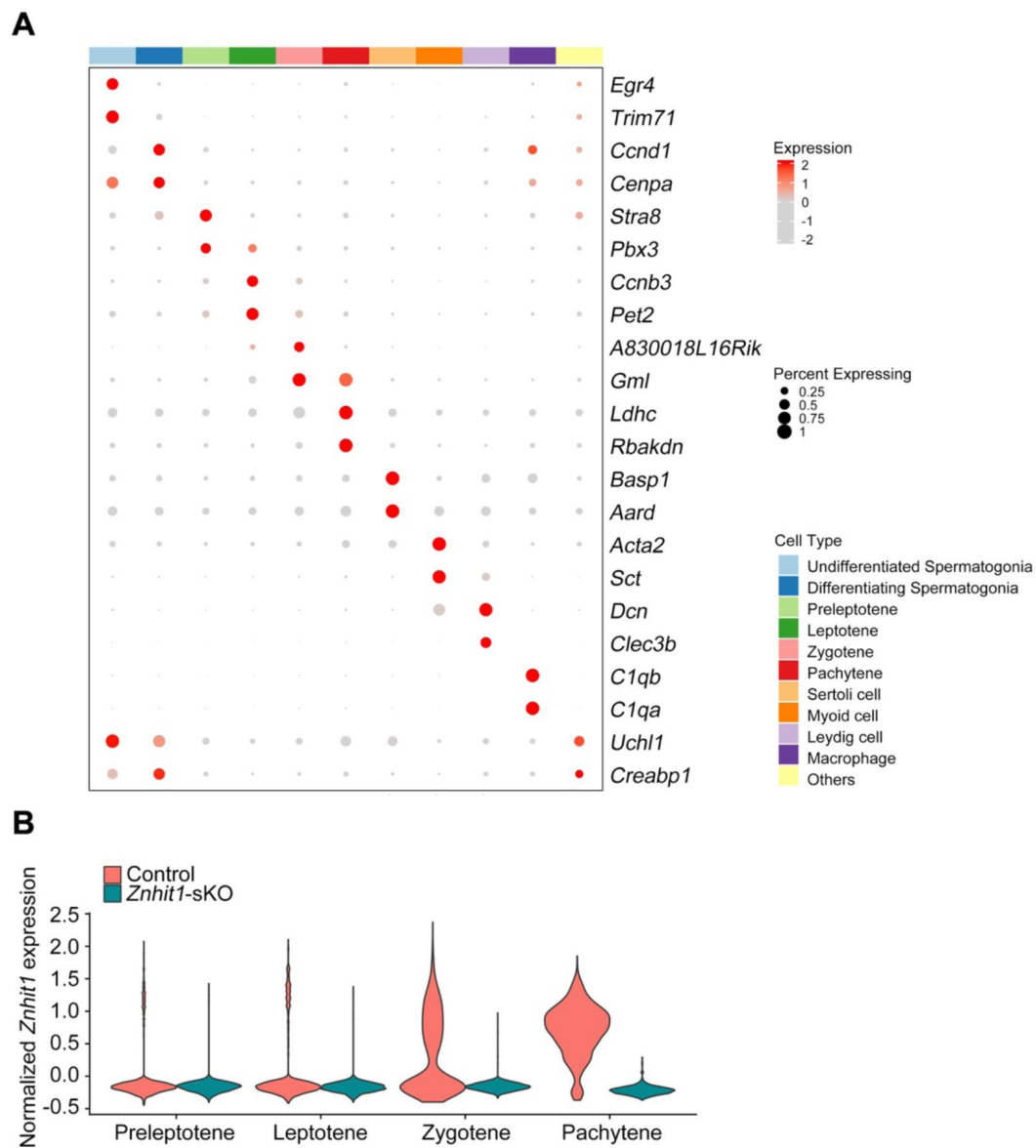


Figure S4.

Single-cell RNA-seq of testicular cells.

(A) Dot plot showing the relative expression and the percentage of cells expressing selected markers across scRNA-seq clusters. (B) Violin plot showing the normalized expression of *Znhit1* in control and *Znhit1*-sKO spermatocytes at consecutive stages.

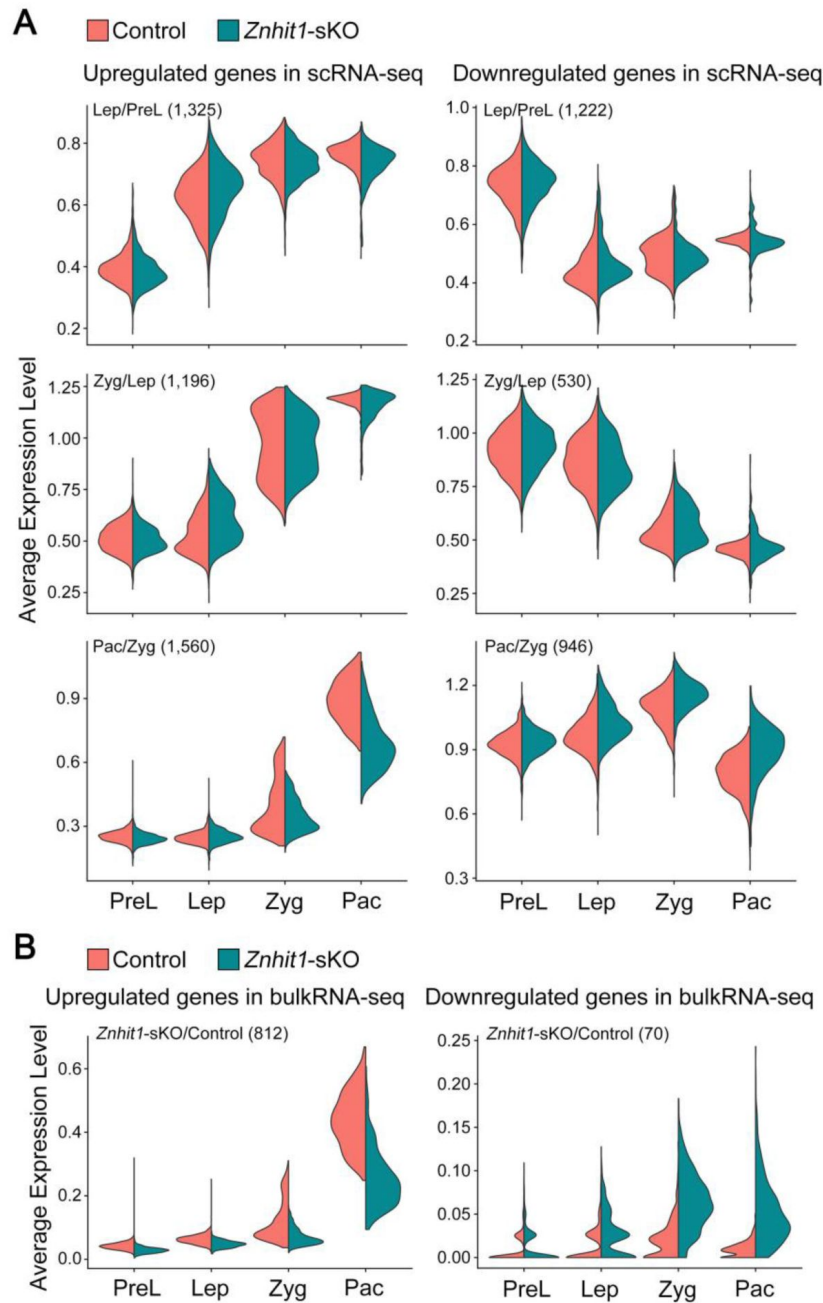


Figure S5.

Single-cell transcriptomic analysis of meiotic genome activation.

(A) Violin plots showing the average expression level of upregulated or downregulated genes between two consecutive spermatocyte stages. (B) Violin plots showing the average expression level of upregulated or downregulated genes between P14 control and *Znhit1*-sKO testicular cells. PreL, preleptotene; Lep, leptotene; Zyg, zygotene; Pac, pachytene.

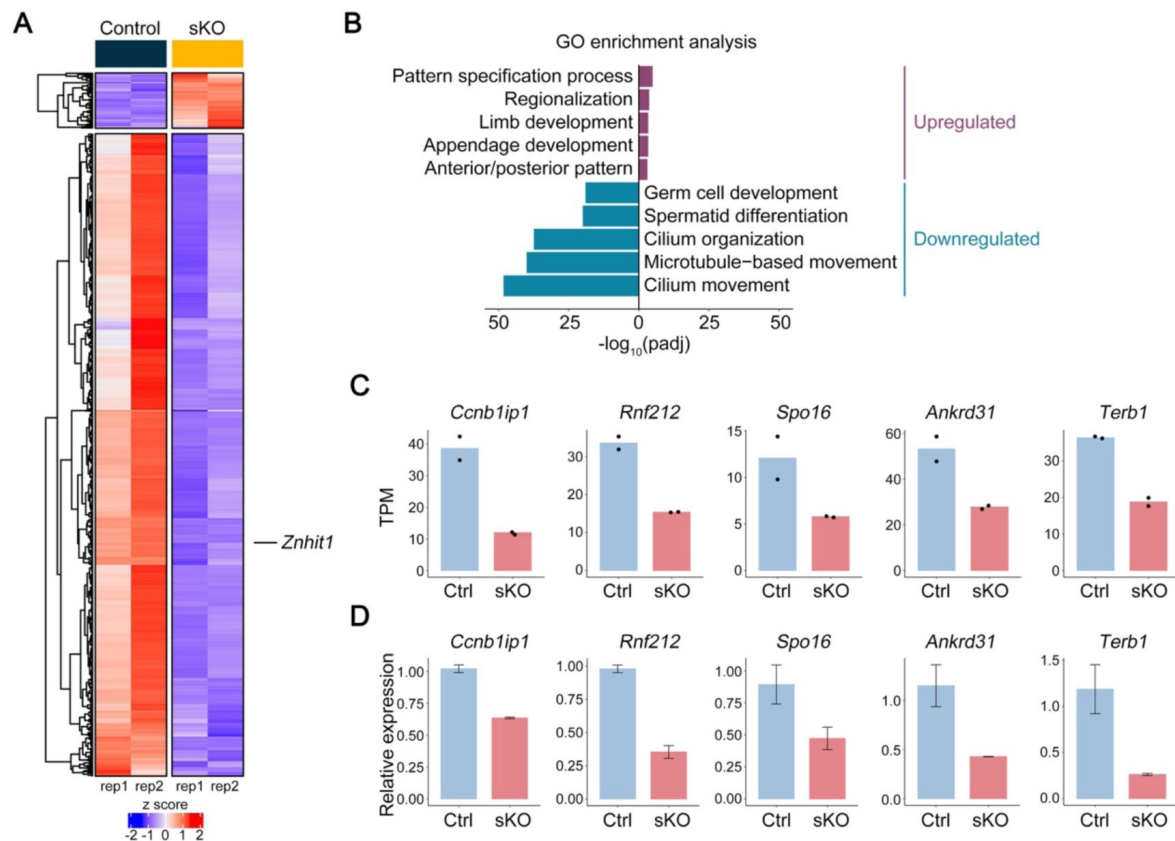


Figure S6.

differentially expressed genes (DEGs) in P14 testicular cells between control and *Znhit1*-sKO mice. **(B)** Gene ontology (GO) analysis of DEGs showing changed biological processes. **(C and D)** Bar charts showing HR-related gene expression in P14 control and *Znhit1*-sKO testes. Data from bulk RNA-seq are shown in **(C)**. Validated data from RT-qPCR are shown in **(D)**. Ctrl, control; sKO, *Znhit1*-sKO.

Figure S8.

***Znhit1* deletion downregulates the expression of A-MYB target genes.**

(A) Violin plot showing the normalized expression of *Mybl1* in control and *Znhit1*-sKO spermatocytes at consecutive stages. (B) Cluster annotation on the basis of the expression of A-MYB target genes in control or *Znhit1*-sKO testicular cells.

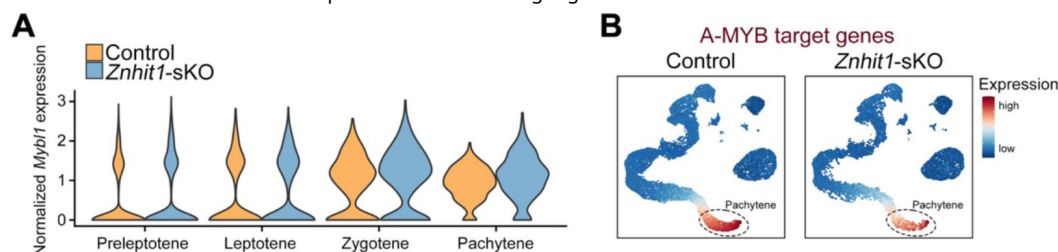
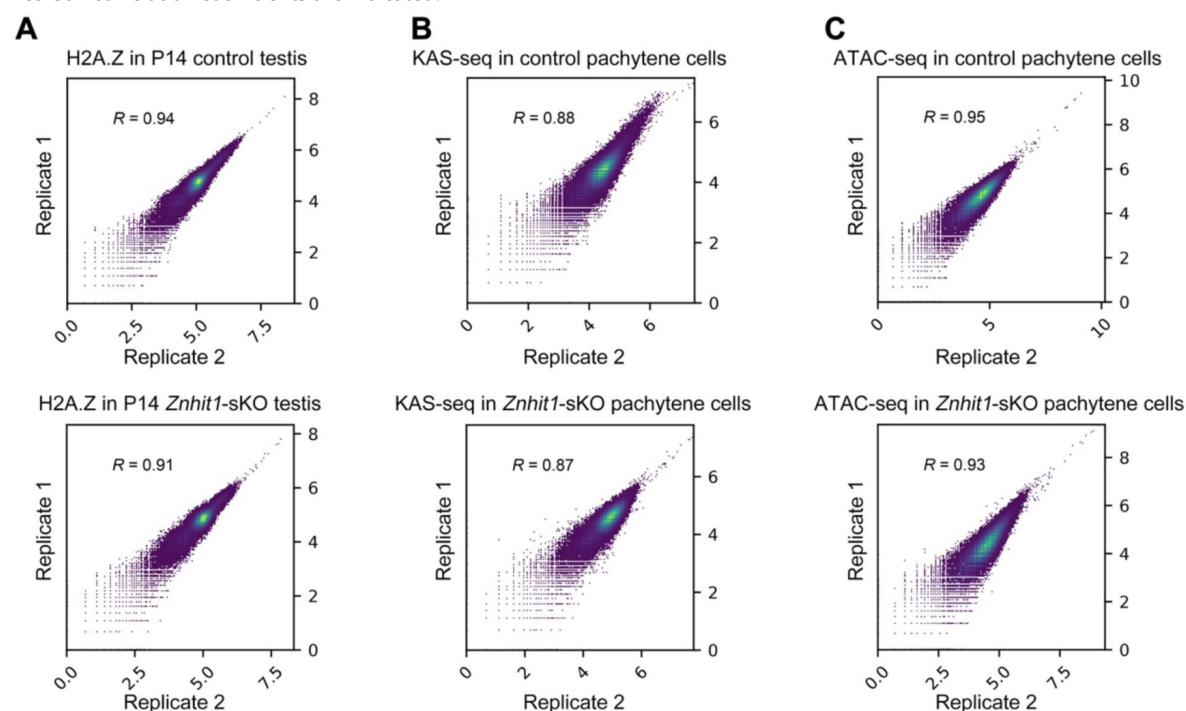


Figure S9.

Reproducibility of replicates in this study.

(A) Scatter plots showing H2A.Z enrichments in P14 control or *Znhit1*-sKO testes for two independent H2A.Z ChIP-seq replicates. Pearson correlation coefficients are indicated. (B) Scatter plots showing KAS-seq signals in P14 control or *Znhit1*-sKO pachytene cells for two independent H2A.Z ChIP-seq replicates. Pearson correlation coefficients are indicated. (C) Scatter plots showing ATAC-seq signals in P14 control or *Znhit1*-sKO pachytene cells for two independent H2A.Z ChIP-seq replicates. Pearson correlation coefficients are indicated.



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Reviewer #1 (Public Review):

Summary:

Sun et al. generated germline-specific cKO mice for the *Znhit1* gene and examined its effect on male meiosis. The authors found that the loss of *Znhit1* affects the transcriptional activation of pachytene. *Znhit1* is a subunit of the SRCAP chromatin remodeling complex and a depositor of H2AZ, and in cKO spermatocytes, H2AZ is not deposited into the gene region. The authors claim that this is why the PGA was not activated. These findings provide important insights into the mechanisms of transcriptional regulation during the meiotic prophase.

Strengths:

The authors used samples from their original mouse model, analyzing both the epigenome and the transcriptome in detail using diverse NGS analyses to gain new insights into PGA. The quality of the results appeared excellent.

Weaknesses:

Overall, the data is inconsistent with the authors' claims and does not support their final conclusions. In addition, the sample used may not be the most suitable for the analysis, but a more suitable sample would dramatically improve the overall quality of the paper.

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Reviewer #2 (Public Review):

Summary:

The study demonstrates that *Znhit1* regulates male meiosis, with deletion causing pachytene failure associated with defective expression of pachytene genes and subtle effects on X-Y pairing and DSB repair. The authors attribute this phenotype to the defective incorporation of the *Znhit1* target H2A.Z into chromatin.

Strengths:

The paper and the figures are well presented and the narrative is clear. Evidence that the conditional deletion strategy removes *Znhit1* is strong, with multiple orthogonal approaches used. Most of the meiotic phenotyping is well performed, and the omics analysis clearly identifies a dramatic effect on the meiotic gene expression program. The link to H2A.Z and A-MYB adds a mechanistic angle to the study.

Weaknesses:

(1) Current literature demonstrates that meiotic mutants arrest at one of two stages: midpachytene (stage IV of the seminiferous cycle) or metaphase I (stage XII of the seminiferous cycle). This study documents that in the *Znhit1* KO the midpachytene marker H1t appears normally, but that cells arrest before diplotene. If this is true, then arrest must occur during late pachytene, which based on my knowledge has never been documented for a meiotic KO. To resolve this, the authors should present stronger histological substaging evidence to support their claim.

(2) The authors overlooked the possible effects of *Znhit1* deletion on MSCI. Defective MSCI is a well-established cause of pachytene arrest. Actually, the fact that they see X-Y pairing failure should alert them even more strongly to this possibility because MSCI failure is often associated with defective X-Y pairing. This could be easily addressed by examination of their RNAseq data.

(3) The recombination assays need attention.

- In the text the authors state that they studied RPA2 and DMC1, but the figures show RPA2 and RAD51.
- The RPA counts are not quantitated.
- The conclusion that crossover formation fails (based on MLH1 staining) is not justified. This marker does not appear in wt males until late pachytene, so if cells in this mutant are dying before that stage, MLH1 cannot be assessed.
- The authors state that gH2AZ persists in the KO, but I'm not convinced that they are comparing equivalent stages in the wt and KO. In Figure 3C, the pachytene cell is late, whereas in the mutant the pachytene cell is early or mid (when residual gH2AX is expected, even in wt males).
- Previous work (PMID: 23824539) has shown that antibodies reportedly detecting pATM in the sex body are non-specific. I therefore advise caution with the data shown in Figure 3D.

(4) RNAseq data. The authors show convincingly that Znhit1 activates genes that are normally upregulated at the zyg-pachytene transition. They should repeat the analysis for genes normally upregulated at the prelep- lep and lep-zyg transition to show that this effect is really pachytene-gene specific.

(5) I am puzzled that the title and overall gist of the study focuses on H2A.Z, when it is Znhit1 that has been deleted.

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Reviewer #3 (Public Review):

Summary:

Sun et al. present a manuscript detailing the phenotypic characterization of loss of Znhit1 in male germ cells. Znhit1 is a subunit of the chromatin regulating complex SRCAP that functions to deposit the histone variant H2A.Z. Given that meiosis, and specifically meiotic recombination, occurs in the context of the dynamic condensing of chromosomes, the role of chromatin regulators in general, and histone variants specifically, in mammalian meiosis is an active area of research. Previous work has shown that H2A.Z is found at the locations of recombination in plants, although H2A.Z was previously not found at recombination sites in mammalian meiosis. Here the authors use a conditional approach to ablate Znhit1 in spermatocytes and characterize a block in meiosis in prophase I in the transition from pachytene to diplotene stage.

Strengths:

The authors combine current methods in immunohistochemistry and functional genomics to provide strong evidence of meiotic block upon the loss of Znhit1. They find that loss of Znhit1 leads to reduced incorporation of the histone variant H2A.Z, specifically at promoters and enhancers. Further, RNA sequencing found more genes are down-regulated upon loss of Znhit1 compared to upregulated, suggesting that incorporation of H2A.Z is critical for the expression of genes necessary for successful meiotic progression.

A strength of the manuscript is tying the locations of changes in H2A.Z deposition with binding of the transcription factor A-MYB, providing a mechanism that can potentially combine the changes in chromatin regulation with variable binding of a transcription factor in gene expression in pachytene stage spermatocytes.

Weaknesses:

A weakness in the single-cell RNA experiment using cells from 16-day-old male mice. The authors suggest that the rationale for the experiment was to determine where the *Znhit1*-sKO mutant showed an arrest in meiosis, and claim that this is the pachytene stage. However, in the 'first wave' of meiosis 16-day-old mice are just beginning to enter pachytene, so cells from later meiotic stages will be largely absent in these tubules. This is clear from the UMAP showing a similar pattern of cell distributions between wild-type and mutant mice. Using older mice would have better demonstrated where the mutant and wild-type mice differ in cell-type composition.

The authors use the term pachytene genome activation (PGS) in the manuscript to suggest a novel process by which genes are specifically increased in expression in the pachytene stage of meiotic prophase I, without reference to literature that establishes the term. If the authors are putting forward a new concept defined by this term, it would strengthen the manuscript to describe it further and delineate what the genes are that are activated and discuss potential mechanisms.

Generally speaking, the authors present solid evidence for a pachytene block in male germ cell development in mice lacking *Znhit1* in spermatocytes. The evidence supporting a change in gene expression during pachytene, that more genes are downregulated in the mutant compared to increased expression, and changes in histone modification dynamics and placement of H2A.Z all support a role in alterations in meiotic gene regulation. However, the support that changes in H2A.Z impacting meiotic recombination (as suggested in the manuscript title) is less supported, rather than a general cell arrest in the pachytene stage leading to cell death. The conclusions around the role of *Znhit1* influencing meiotic recombination directly could use further justification or mechanistic hypothesis.

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